

Phishing Website Detection using Machine Learning

Final Project – Data Science in Cybersecurity

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Executive Summary:

This project evaluates a published machine learning solution for phishing website detection and checks whether the author's claims are supported by the results.

The original project uses URL and website features to classify websites as phishing or legitimate. The dataset contains 10,000 websites, including 5,000 phishing websites and 5,000 legitimate websites. A total of 17 features were extracted from each website and used to train the models.

In this project, I implemented and evaluated three machine learning models: Logistic Regression, Decision Tree, and Random Forest. The models were trained using an 80/20 train-test split and evaluated using Accuracy, Precision, Recall, F1-score, and the Confusion Matrix.

The results showed that Random Forest gave the best performance, followed by Decision Tree. Logistic Regression had a lower recall, which means it missed more phishing websites.

Overall, the results support the author's main claims. However, the project also has some limitations, such as using handcrafted features and testing the models on only one dataset.

Overall, this study showed that machine learning is an effective way to detect phishing websites, and that tree-based models performed the best for this task.

Summary of the Source:

➤ **Problem Description :**

Phishing is one of the most common cyberattacks used to steal personal information such as usernames, passwords, and financial data. Attackers create fake websites that look like real ones to trick users into giving their information.

As phishing attacks become more common and advanced, automatically detecting phishing websites has become an important task in cybersecurity.

➤ **Why the Problem is Important? :**

Phishing attacks can lead to financial loss, identity theft, and unauthorized access to personal or company accounts. Detecting phishing websites before users visit them helps reduce these risks and improve cybersecurity.

➤ **The proposed solution:**

The selected project uses machine learning to detect phishing websites. Instead of using manually created rules, it uses URL and website features to train models that can classify websites as either phishing or legitimate.

➤ **The dataset used:**

The dataset contains 10,000 websites, including 5,000 phishing websites from PhishTank and 5,000 legitimate websites from the University of New Brunswick dataset. For each website, 17 features were extracted from the URL, domain information, and website content.

➤ **The model or methodology employed:**

The original project extracted 17 features from each website and used them to train different machine learning models for phishing detection.

In this project, I implemented and evaluated three models: Logistic Regression, Decision Tree, and Random Forest. The models were trained using an 80/20 train-test split and evaluated using Accuracy, Precision, Recall, F1-score, and the Confusion Matrix.

Critical Evaluation:

➤ **The main claims:**

The author claims that machine learning models can effectively detect phishing websites using URL and website features.

The original project also shows that tree-based models, especially XGBoost, achieve the best performance for phishing website detection.

➤ **Are the claims supported by the presented evidence? :**

The results support the author's main claim. In this project, all three machine learning models were able to detect phishing websites with good accuracy.

Although I did not implement all the models used in the original project, the results from Logistic Regression, Decision Tree, and Random Forest show that machine learning is an effective way to detect phishing websites.

Among the three models, Random Forest achieved the best performance, while the Decision Tree also gave good results.

➤ **Evaluation Methodology:**

The evaluation method is suitable for this classification task. The dataset was split into training and testing sets using an 80/20 ratio, so the models were tested on unseen data.

The models were evaluated using Accuracy, Precision, Recall, F1-score, and the Confusion Matrix. These metrics give a better evaluation than accuracy alone because phishing detection needs to balance false positives and false negatives.

➤ **Limitations:**

The project also has some limitations.

First, the dataset contains only 10,000 websites, so it may not represent all types of phishing websites found on the Internet.

Second, the features are manually designed, so they may not capture all new phishing attack patterns.

Finally, the models were evaluated using only one train-test split. Using methods like cross-validation could give a more reliable evaluation of the models.

➤ **Are the conclusions justified?**

Overall, the results support the author's conclusions.

My results also showed that the tree-based models performed better than Logistic Regression on this dataset. Although I evaluated fewer models than the original project, I reached a similar conclusion: machine learning is an effective way to detect phishing websites, and Random Forest gave the best results among the models I tested.

Feature Engineering Analysis:

➤ **feature engineering performed:**

The original project extracted 17 features from each website. These features were taken from the URL, domain information, and the website content.

Instead of using the raw URL, each website was represented by a set of numerical features that were used to train the machine learning models.

➤ **Features Used:**

The extracted features are divided into three main groups:

- Address bar features, such as URL length, IP address, the '@' symbol, URL depth, URL redirection, TinyURL, and prefix or suffix in the domain.
- Domain features, such as DNS record, web traffic, domain age, and domain expiration time.
- HTML and JavaScript features, such as iframe usage, mouse-over behavior, disabled right-click, and website forwarding.

These features were chosen because they are common characteristics of phishing websites.

➤ **Data Transformations:**

Most of the features are binary, with values of 0 or 1, while a few features, such as URL depth, are numerical.

All the features were already in numerical form, so no encoding was needed. The dataset also had no missing values, so no imputation was required.

The project did not use feature scaling or dimensionality reduction. This is acceptable because Decision Tree and Random Forest do not require normalized features.

➤ **The redundancy in the system:**

Some features may provide similar information. For example, URL length, URL depth, and TinyURL usage are all related to the URL structure.

Also, domain age and domain expiration time both describe the website domain and may contain similar information.

This can be checked using correlation analysis or feature importance. If some features are very similar, they can be removed without greatly affecting the model's performance.

➤ **Was the Feature Engineering Meaningful:**

Overall, the feature engineering process was useful because the selected features are related to common phishing techniques.

Phishing websites often use long URLs, shortened links, suspicious domain names, or misleading HTML elements to trick users. The extracted features capture many of these characteristics.

By converting each website into numerical features, the machine learning models can learn these patterns and classify websites as phishing or legitimate.

➤ **Additional features that could potentially improve performance:**

The selected features provide useful information, but adding more features could improve the model.

For example, features such as the number of subdomains, suspicious words in the URL, SSL certificate validity, domain reputation, WHOIS information, website similarity, and URL analysis could be added.

These features could help the model detect more advanced phishing websites.

Reproducibility Analysis:

➤ **Code Execution:**

The original GitHub repository included the code and dataset needed to reproduce the experiments.

The code was run successfully in Google Colab, and all three machine learning models were trained without any major problems.

➤ **Are all required files and dependencies are available:**

The repository included the dataset and all the files needed to reproduce the experiments.

The implementation uses common Python libraries such as pandas, numpy, matplotlib, and scikit-learn. These libraries are easy to install and widely used.

➤ **Hidden Preprocessing:**

No major hidden preprocessing steps were found.

The preprocessing process is clearly described. It includes loading the dataset, checking for missing values, selecting the input features, and splitting the data into training and testing sets.

Because these steps are simple and well documented, the experiments can be reproduced easily.

➤ **The overall reproducibility of the work:**

Overall, the project is reproducible. The dataset, source code, and implementation details are available, so other people can repeat the experiments and get similar results.

The exact results may be slightly different because of software versions or random initialization, but the overall conclusions should stay the same.

Experimental Results:

➤ **Experimental Setup:**

The experiments were performed using the phishing website dataset, which contains 10,000 websites. The dataset was split into training and testing sets using an 80/20 ratio. Three machine learning models were implemented and evaluated:

- Logistic Regression
- Decision Tree
- Random Forest

All three models were trained on the same training data and tested on the same test set to make the comparison fair.

➤ **Modifications introduced:**

The models were evaluated using Accuracy, Precision, Recall, F1-score, and the Confusion Matrix.

- Accuracy shows the overall percentage of correctly classified websites.
- Precision shows how many websites predicted as phishing were actually phishing websites.
- Recall shows how many phishing websites were correctly detected.
- F1-score combines precision and recall into a single measure.
- The Confusion Matrix shows the number of correctly and incorrectly classified phishing and legitimate websites.

➤ **The Models You Trained:**

Three machine learning models were trained and evaluated:

- Logistic Regression
- Decision Tree
- Random Forest

All three models were trained on the same training set and tested on the same test set to make the comparison fair.

➤ **The evaluation metrics:**

The following evaluation metrics were used:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

These metrics give a better evaluation than accuracy alone because they measure both the correctness of the predictions and how well the models detect phishing websites.

➤ **The obtained results:**

The results show that Random Forest achieved the best performance, with an accuracy of 86.7% and the highest F1-score. The Decision Tree also gave very good results and performed better than Logistic Regression. Logistic Regression had the lowest recall, which means it missed more phishing websites.

Overall, Random Forest gave the best balance between precision and recall and was the best model in this project.

Error Analysis:

➤ **Examples of Model Failures:**

The models were not able to classify every website correctly. Some phishing websites were classified as legitimate, and some legitimate websites were classified as phishing. These mistakes show that no model was perfect, even though Random Forest gave the best overall performance.

➤ **Patterns in the Errors:**

The results show that Logistic Regression missed more phishing websites than the tree-based models because it had the lowest recall.

Decision Tree and Random Forest missed fewer phishing websites, showing that they were better at learning the patterns in the data.

➤ **Cybersecurity Implications of the Errors:**

In phishing website detection, False Negatives are usually more dangerous than False Positives.

A False Negative means that a phishing website is classified as legitimate, so users may visit a malicious website. This can lead to stolen passwords, financial loss, or identity theft.

A False Positive means that a legitimate website is classified as phishing. This may be inconvenient for users, but it is usually less harmful than missing a phishing website.

➤ **The Trade-off Between False Positives and False Negatives:**

A good phishing detection model should reduce both False Positives and False Negatives. However, improving one type of error can sometimes increase the other. For phishing detection, reducing False Negatives is usually more important because missing a phishing website can have serious consequences.

Among the three models, Random Forest gave the best balance between detecting phishing websites and reducing false alarms, making it the best model for this task.

Conclusions:

➤ **Key findings:**

In this project, I trained and evaluated three machine learning models for phishing website detection: Logistic Regression, Decision Tree, and Random Forest.

The results showed that Random Forest gave the best performance, followed by Decision Tree. Logistic Regression worked as a good baseline model, but its overall performance was lower than the other two models.

➤ **Lessons learned:**

This project showed that it is important to evaluate machine learning models using different metrics, not just accuracy. Precision, recall, F1-score, and the confusion matrix helped show the strengths and weaknesses of each model.

The project also showed that features extracted from the URL and website can be used to effectively detect phishing websites.

➤ **Strengths and weaknesses of the proposed solution:**

One strength of this solution is that it uses simple features that are easy to extract while still giving good results.

However, the solution also has some limitations. It uses manually designed features and was tested on only one dataset. In addition, Logistic Regression missed many phishing websites, giving it a lower recall than the tree-based models.

➤ **Suggestions for future improvements:**

Future work could test more machine learning models, such as XGBoost or LightGBM, and tune their parameters to improve performance.

Another improvement would be to add more URL and website features or test the models on newer phishing datasets to see how well they generalize.

Summing It Up:

The selected project focuses on phishing website detection using machine learning. It uses URL and website features to classify websites as phishing or legitimate.

In this reproduction study, I trained and evaluated three machine learning models: Logistic Regression, Decision Tree, and Random Forest.

The results support the author's main claim that machine learning can effectively detect phishing websites. Among the three models, Random Forest gave the best performance. The selected features were useful because they captured common characteristics of phishing websites. The evaluation was also appropriate because several evaluation metrics were used to measure the models' performance.

Although the models achieved good results, the project has some limitations. It uses handcrafted features and was tested on only one dataset. In the future, more datasets, additional models such as XGBoost or LightGBM, and hyperparameter tuning could be used to improve the results.

Overall, this reproduction study confirmed the author's main claims and showed that this is an effective approach for phishing website detection.